

Antibiotics

Cell Wall Inhibitors



Edited by : أسيل السلمي



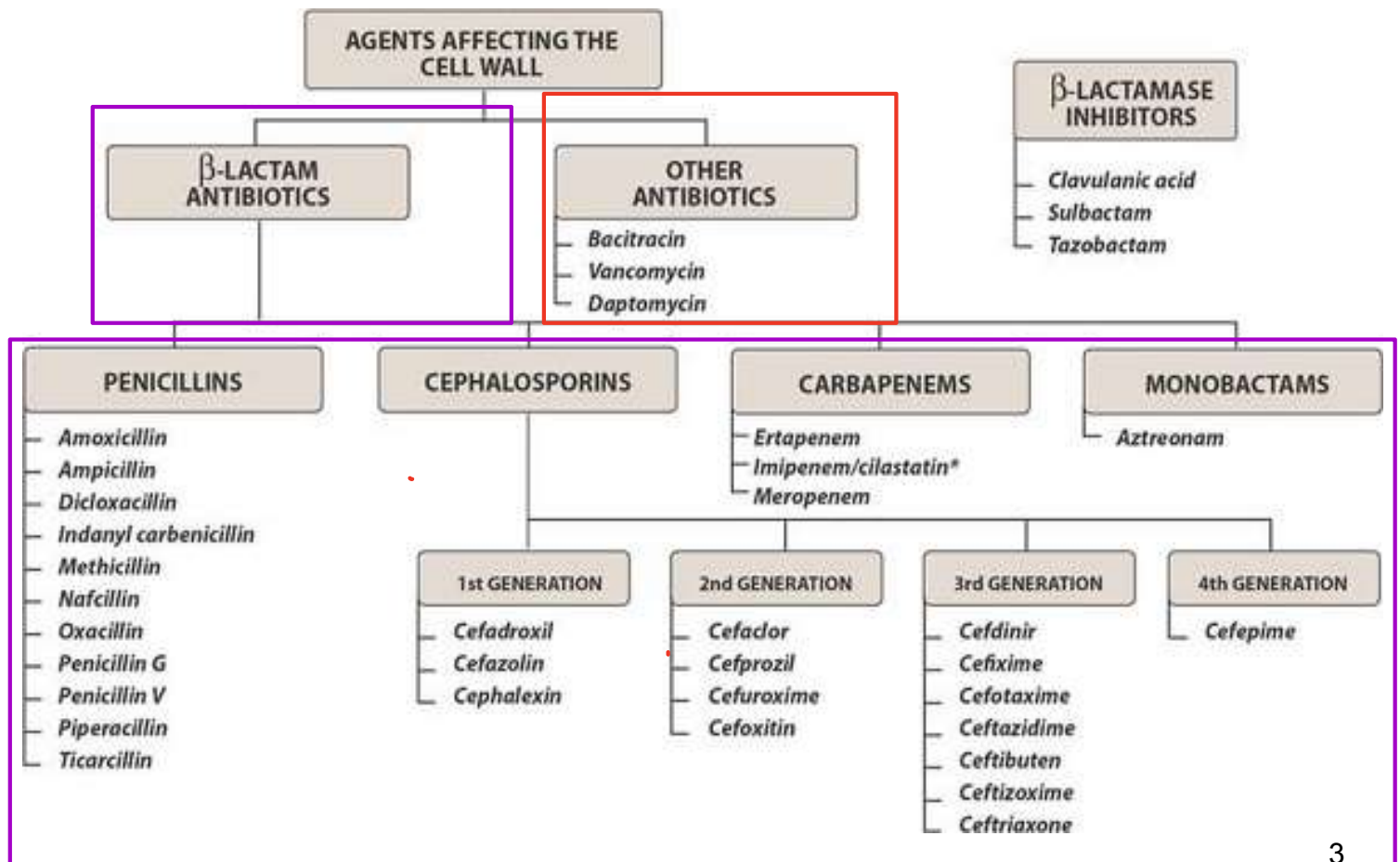
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Intended learning outcomes (ILOs)

- How do cell wall inhibitors work?
- Discuss their pharmacokinetics
- Apply the pharmacological effects of antimicrobial drugs for its major clinical uses and discuss their adverse effects
- Define the actions and uses of **Vancomycin**
- **Why do you always have to finish your prescription?**

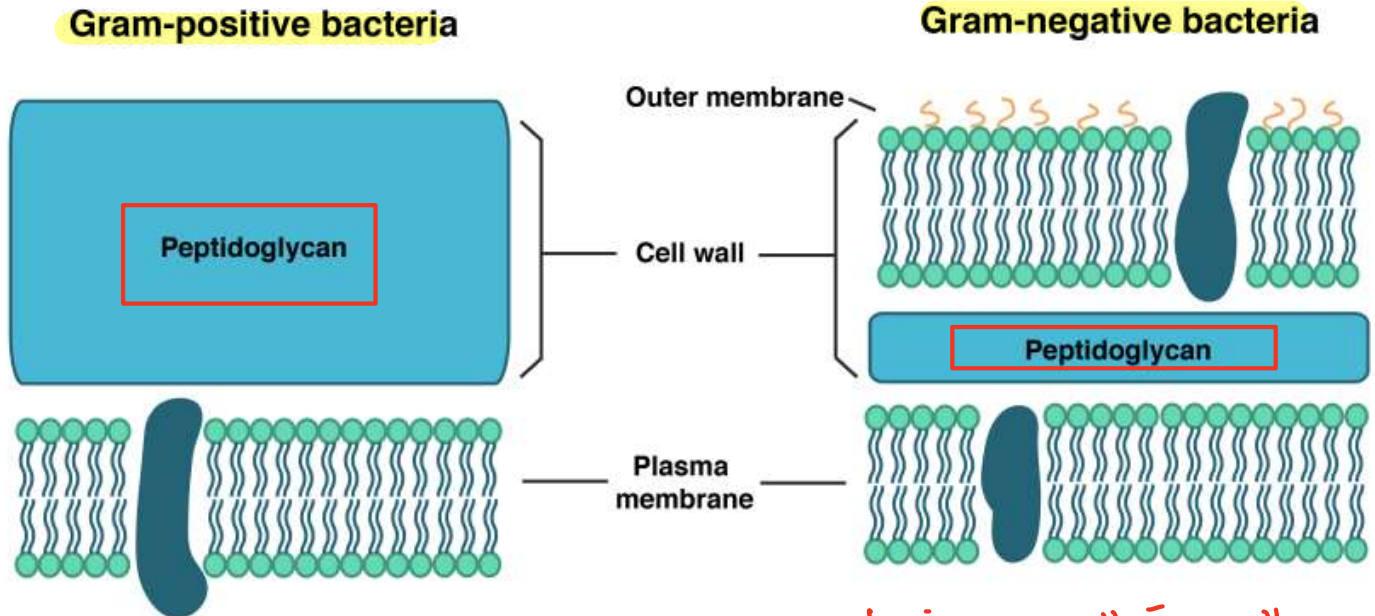


Bacterial cell wall structure

more Resistance

عملية الدخول صعبة

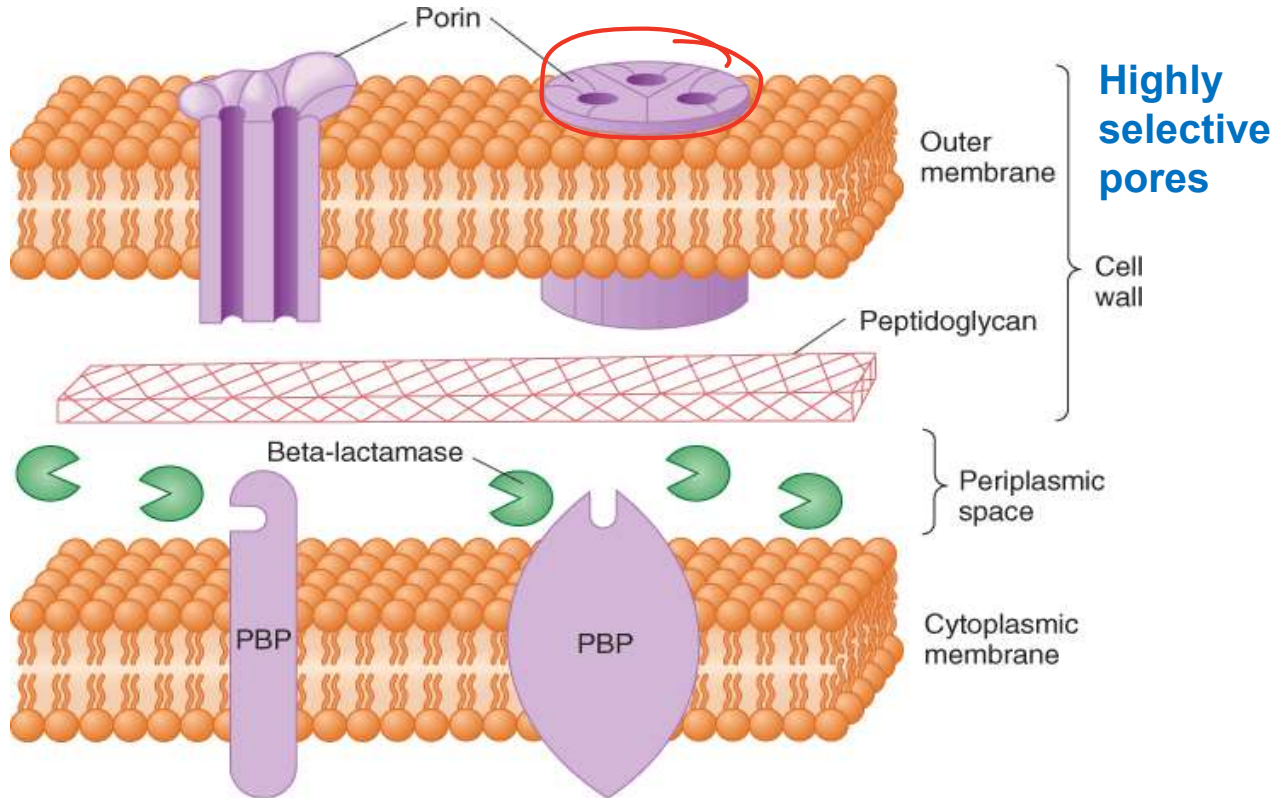
سهلة



الدخول الى بدنها
فأبدا تستهدف - G

Bacterial cell wall structure

بسی تر هیچ لو جعل
مقاومة ید تغییر سده
ار Poirn ر ما حیدر یدخل



Source: Trevor AJ, Katzung BG, Masters SB: *Pharmacology Examination & Board Review*, 9th Edition: www.accesspharmacy.com
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PBP: penicillin binding peptide

Resistance against Penicillins

Natural (Innate Resistance)

Lack a peptidoglycan cell wall

- Fungi
- Virus
- Human cells
- Mycoplasma
- Mycobacteria
- Chlamydia

Has peptidoglycan cell wall but still resistant

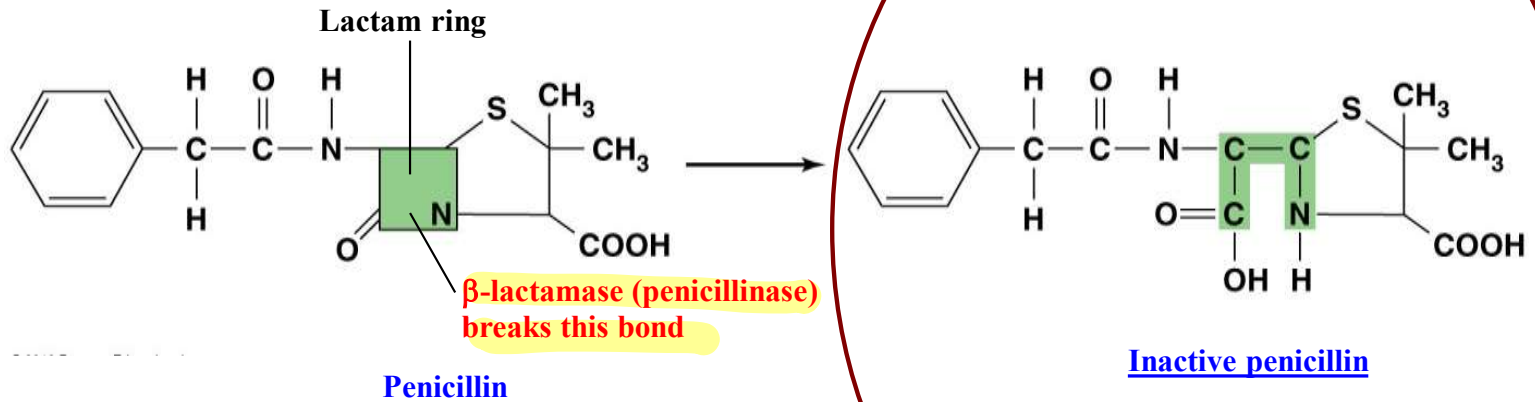
- *P. aeruginosa* ≠ PenG



Acquired

(Initially susceptible then later got resistant)

- 1- β -lactamase production
(plasmid-mediated β -lactamase)
- 2- Altering the porin ($\downarrow$ permeability to drug)
- 3- Increase efflux pump
- 4- Alteration of PBP (MRSA, VRSA)
(*S. pneumoniae* ≠ PenG)



How beta-lactamase (penicillinase) renders penicillin inactive

لو انكسر حلقة بيتا بيريميد البنسلين inactive

الحل؟؟

نضيف مواد خاملة شبيهة حلقة بيتا نخدع البكتيريا

Antibiotics acting on cell wall of bacteria

- Beta lactams:

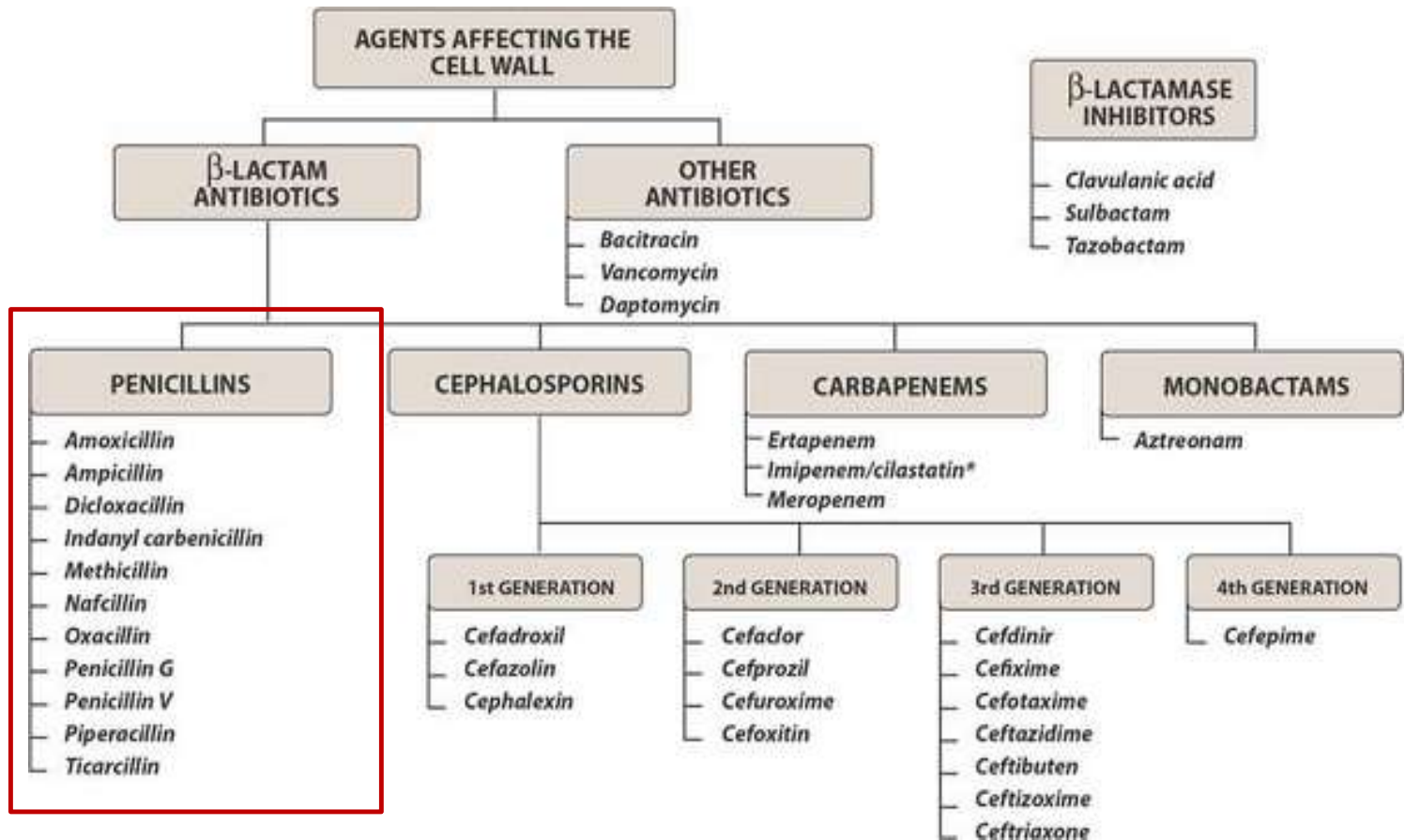
(Penicillins, cephalosporins, carbapenems, monobactam)

طغفة بيتا

- Glycopeptides: بروند طغفة بيتا

(Vancomycin, teicoplanin)

Bacterial cell wall synthesis inhibitors



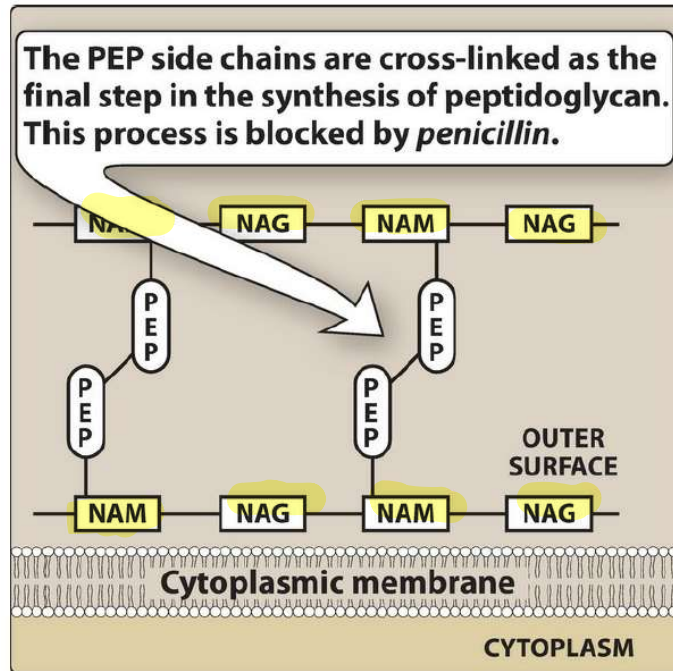
Penicillins: Mechanism of Action

Penicillins enter the bacteria via the cell wall

- Inside the cell, they bind to penicillin-binding-protein (PBP)
✱ (transpeptidase enzyme responsible for peptidoglycan cross-linkings)
- Once bound, normal cell wall synthesis is disrupted.
- Bacteria become hyperosmotic/ love to absorb water
- Bacteria swelling then burst.
- These drugs are bactericidal. قتل بكتريا
- Active against actively proliferating and not resting bacteria.
- Inactive against protozoa, mycobacteria, fungi and viruses.



Penicillins: Mechanism of Action



PEP: cross-linking peptides

PENICILLINS

MRSA, Pseudomonas
اي دواء يجيبهم مهم

Natural Pens.

v.v. NaRRow spectrum

Extended-spectrum Pens.

Antipseudomonal Pens.

(Benzyle Penicillins)

(Antistaphylococcal Pens.)

(Aminopenicillins)

(Antipseudomonal Pens.)

Pen G

Pen V

I.V

P.O.

Methicillin
Cloxacillin
Floxacillin
Oxacillin
Dicloxacillin
Nafcillin

Ampicillin (I.V)
Amoxicillin (P.O.)
Ampicillin +
Sulbactam = Unasyn
Amoxicillin +
clavulanic acid
= Augmentin

Carbenicillin
Ticarcillin +
clavulanic acid =
Timentin
Piperacillin +
tazobactam = Zosyn

- G+ cocci
- G+ bacilli
- G- cocci (Gonococcus)
- Spirochetes (T. pallidum) (Syphilis)

Staphylococci
عشان عذا اما بنسجولها
دا يا

G+ قالة
بشكل كام

مضروب الطول ١.٨

نستخومها للان

Procaine penicillin G

Benzathine Penicillin G



هذا القروب كمان يغير له Hydrolysis
و يحتاج combination therapy

	PENICILLIN G	ANTI-STAPH PENICILLINS	EXTENDED SPECTRUM PENICILLINS	ANTI- PSEUDOMONAL PENICILLINS
		Cloxacillin Flucloxacillin Nafcillin	Ampicillin Amoxicillin	Ticarcillin Azlocillin Pipracillin
Spectrum	Gram +ve spp. and few Gram -ve cocci <i>T. pallidum</i>	<i>Staphylococci</i>	Gram +ve and Gram -ve spp. (except <i>Pseudomonas</i>)	<i>Pseudomonas</i> and Gram -ve spp.
β-lactamase susceptibility	Yes	No	Yes	Yes
			They are combined with clavulanic acid or sulbactam to cover β-lactamase producing bacteria.	

الحالات
النادرة

EXAM

- Long-acting penicillins:
- Some Penicillins are used as **Depot form (I.M)**; active penicillins release slowly (Given in upper outer part of Buttocks, children; thigh)

ناخذه
١٢ اليوم Procaine Pencillin G (12-24 h) ١٢
ناخذه
١٢ الشهر Benzathine Penicillin G (Rheumatic fever) (3-4 Weeks) * *

Used as Prophylaxis:

- 1- Against streptococcus pyogens in patients with Rheumatic fever * *
- 2- Patients with artificial valves (endocarditis)
- 3- After delivery (newborn with gonorrheal conjunctivitis)

EXAM

مراجعة الـ قبل
- بعض أنواع البنسلين اميف مادية
خاملة اذرع فيها البكتريا

β - Lactamase Inhibitors

Formulations with B-lactamase inhibitors protect amoxicillin or ampicillin from enzymatic hydrolysis and extends their antimicrobial spectrum.:

★ Clavulanic acid + amoxicillin (**Augmentin**) – Oral.

★ Sulbactam + ampicillin (**Unasyn**) (IV-IM).





بكتريا متحركة واي دوائ
 به جديدها احفظله

Pseudomonas Euroginosa

Penicillins: Pharmacokinetics

➤ Absorption:

- Most of the penicillins are poorly absorbed orally Except:

Penicillin V

Amoxicillin

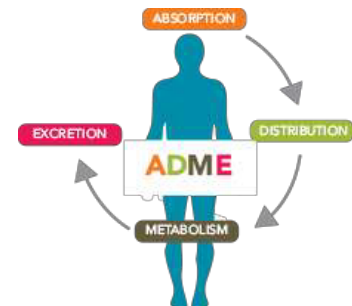
Amoxicillin + clavulanic acid = Augmentin®

Indanyl Carbenicillin

- Given in empty stomach (1 h before or 2 h after food)

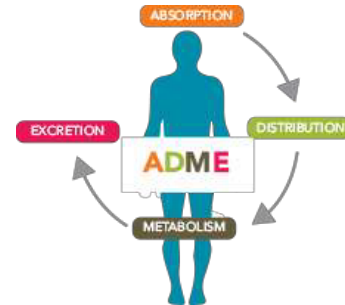
- Amoxicillin is absorbed mostly from GIT
- Ampicillin is very poor oral absorption

↳ IV or IM



➤ Distribution:

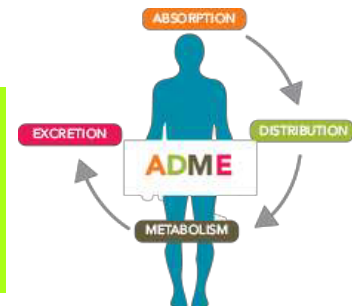
- **Not** lipid soluble to pass through cell membrane; not good for intracellular bacteria.
- **Penicillins normally can not cross BBB (Ionized molecules)**; only in inflammation (e.g. Meningitis) Penicillins can cross into CSF and cross bone (osteomyelitis)
- If cross Placenta- **not** teratogenic
- **The safest Antibiotics to be given to pregnant women**



➤ Metabolism:

- They are very safe drugs, no need for monitoring. kidney →
 - Excreted mainly unchanged in urine by active tubular secretion (90%) (except. Nafcillin, cloxacillin, dicloxacillin and oxacillin; through liver)
 - $T_{1/2} = 30-60 \text{ min}$ (10 h in renal failure)
-
- **Probenecid** bind with organic acid transport mechanism in the kidney and inhibits Penicillins tubular secretion, can increase their blood levels
 - Penicillins appear in milk
 - Excreted in saliva

قالت ال probenecid بس للفهم وما تعرف
إذا حيي او لا
بس احتياط اعرفو أسم الدواء و الميكانيزم حقه
ومين يزيد ومين السبب



➤ Penicillins : clinical uses

- Streptococcal infections (throat infections, bacterial endocarditis *in patients having dental surgery*)
- Staphylococcal Infections
- Spirochetes (Syphilis) and Gonorrhoea (Gram G- cocci)
- Rheumatic Fever (Benzathin Pen G)
- Diphtheria
- Salmonella infections

- *Penicillins + Aminoglycosides = Synergistic effects*
- Prophylactic - in patients with congenital heart disease
 - Rheumatic fever

EXAM



➤ Adverse effects

1. Type-1 hypersensitivity reaction (IgE-immediate reactions) (any immune reaction which is damaging the tissues (occur in 5-10% of patients))
 - Most dangerous is anaphylactic shock (1-3:10.000 person)
low BP
2. Diarrhoea (Pseudomembranous colitis by *c.difficile* especially with Ampicillin) : Treated with Broad Spectrum
 - Metronidazole orally or Vancomycin orally too. *lies ans*
3. Hypovitaminosis (Vit K)
4. Platelete dysfunction (especially with Piperacillin, Nafcillin)
5. May cause seizures (Not used intrathecally); also inhibit GABA especially procaine penicillin

Very Rare

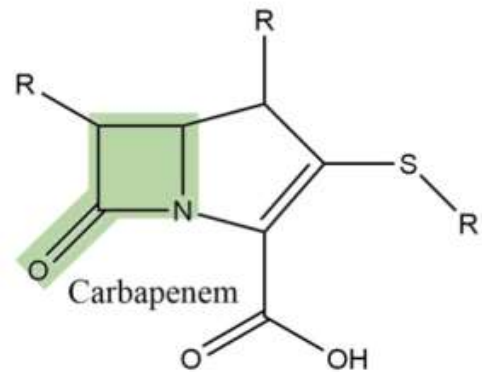
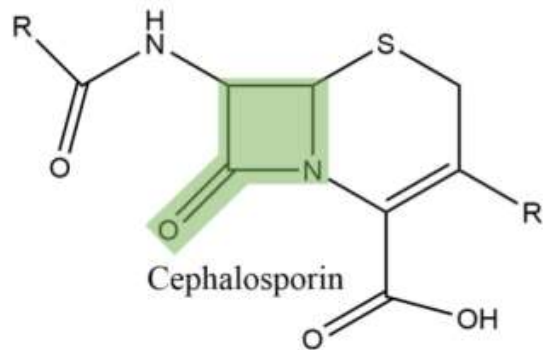
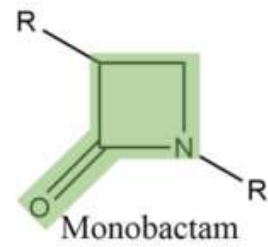
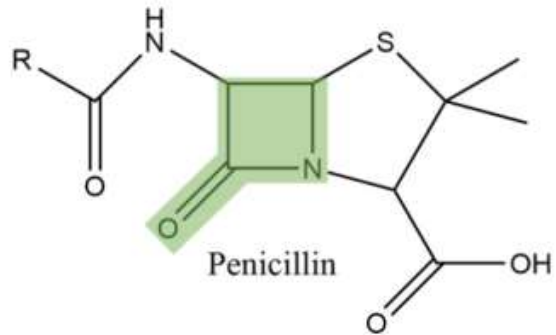


Antibiotic skin testing (AST)

- For identifying patients with IgE-mediated hypersensitivity reactions to antibiotics.
- **About 5-10% hypersensitivity**
- **1:10,000 patients (anaphylactic shock)**
- Performed on the **volar forearm** using the **intradermal method**.
- The injection amounts varied among the hospitals, **ranging from 10 µl**
- The time lag (**15 to 20 minutes**).
- **Wheal and erythema standards for a positive** response varied among the hospitals. (due to degranulation of local mast cell at site of injection)



✓ Positive skin test for specific IgE (> 3-5 mm diameter)

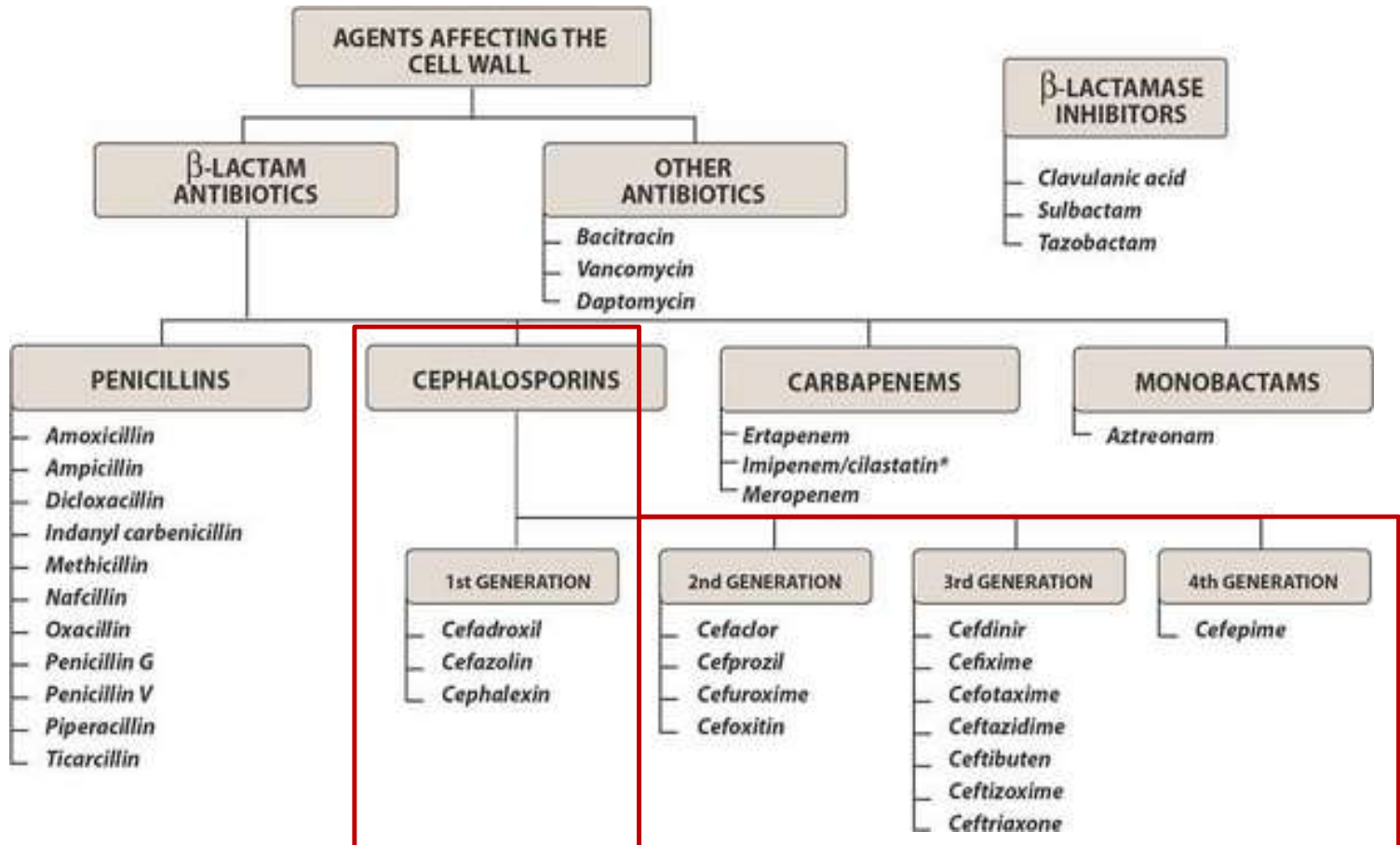




Assume complete
cross-allergenicity
between individual
penicillins.

Cross-sensitivity; occurs when an allergy to one drug causes a reaction to another with a similar structure, such as Penicillins and Cephalosporins.

Bacterial cell wall synthesis inhibitors



Cephalosporins (1945)

- β -lactam antibiotics (similar to penicillins)- (bind to PBP*)
- Act by bacterial cell wall synthesis inhibition

1. More resistant to β -lactamase.

2. Broad spectrum

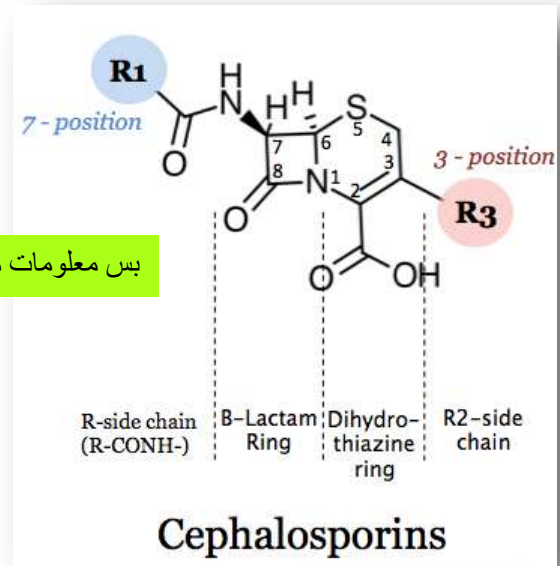
- Bactericidal

- Inactive against: "LAME + C":

- *Listeria monocytogenes*
- *Legionella*
- Atypical (*Chlamydia*, *Mycoplasma*)
- MRSA
- Enterococci
- *Clostridium difficile*



بس معلومات مو للحفظ



- They are commonly used in surgical procedures to reduce post operative infections risk.

Cephalosporins Subgroups

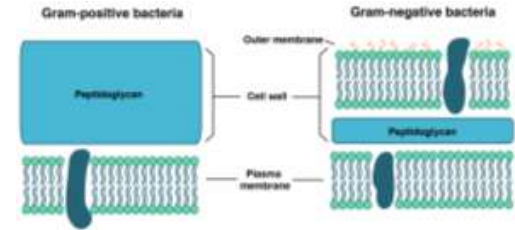
❖ FIRST-GENERATION

حفظ ✖ ✖

- Cefazolin

- Cephalexin (p.o.)

- Cephadrine



- Active against G⁺ organisms (S.aureus, S.pneumoniae, S.pyogenes, S.epidermidis)

- Active against streptococcal pharyngitis

- Commonly used in surgical prophylaxis

- None enter CNS ✖ ✖ ما استخدمها للمeningitis

- Moderate activity against G⁻ organisms (E.coli and K. pneumoniae,...)

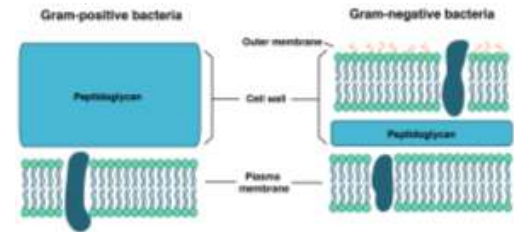


Streptococcal pharyngitis

❖ SECOND-GENERATION

حفظا

- Cefaclor
- Cefuroxime
- Cefoxitin
- Cefprozil



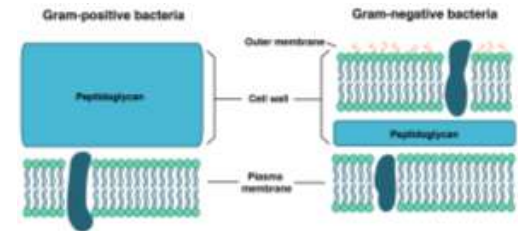
- **Mainly effective against G- organisms.**
- (H.influenza, Moraxella catarrhalis, S. pneumoniae)
- Active against intestine anaerobes (Bacteroides fragilis)
- ✗ **Not active against P.aeruginosae**
- Drugs in this group are used mainly for URTIs (acute otitis media, sinusitis, laryngitis) and LRTIs (chronic bronchitis, COPD)
- **Do not enter CNS**
- **Moderate activity against G+ organisms.**



acute otitis media

❖ THIRD-GENERATION

- Cefotaxime
- Ceftriaxone
- Ceftazidime
- Cefoperazone
- Cefixime (p.o.)



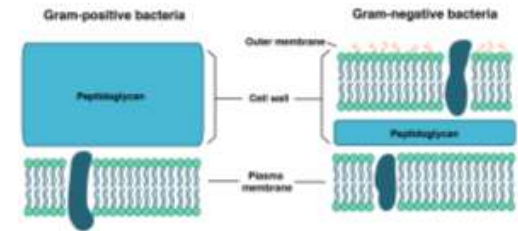
- **Most** active against **G-** organisms (H.influenza, **N. gonorrhea**, E.coli, **N.meningitidis**, M. catarrhalis, Klebsiella spp and **P.aeruginosae**).
- **They penetrate BBB** except (Cefixime and Cefoperazone).
- Less active against G+ organisms



❖ FOURTH-GENERATION

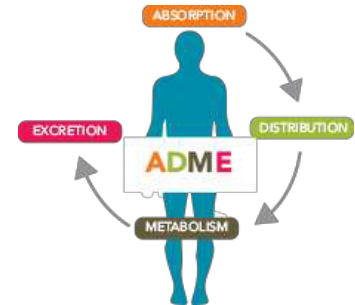
Cefipime (IV)

- Wider spectrum that includes G+ activity and G- cocci activity, plus many G- rods.
- Is active against P. aeruginosae
- They penetrate BBB
- Resistant to most beta-lactamases



Cephalosporins: Pharmacokinetics

- Cephalosporins are **mostly administered parenterally** but some are available for oral use.
- **First and Second** generations **CANNOT** cross **BBB**, however, **Third and Fourth CAN** cross (except **cefoperazone**)
- Relatively lipid insoluble (like penicillins)
- Major elimination is **renal excretion**.
- $T_{1/2} = 30-90$ min.



- **Ceftriaxone has long $T_{1/2} = 4-7$ h**

- **Probenecid** inhibits their elimination and increase their blood levels (except. Ceftazidime and Cefoperazone)

Cephalosporins: clinical uses

M: Meningitis

O: Osteomyelitis

S: Skin infection

T: Typhoid Fever

U: UTI

R: RTI

G: Gonorrhea

E: ENT infection

N: Intestinal Infections

T: Topical

➤ Adverse effects

- Similar to Penicillins ✂ ✂

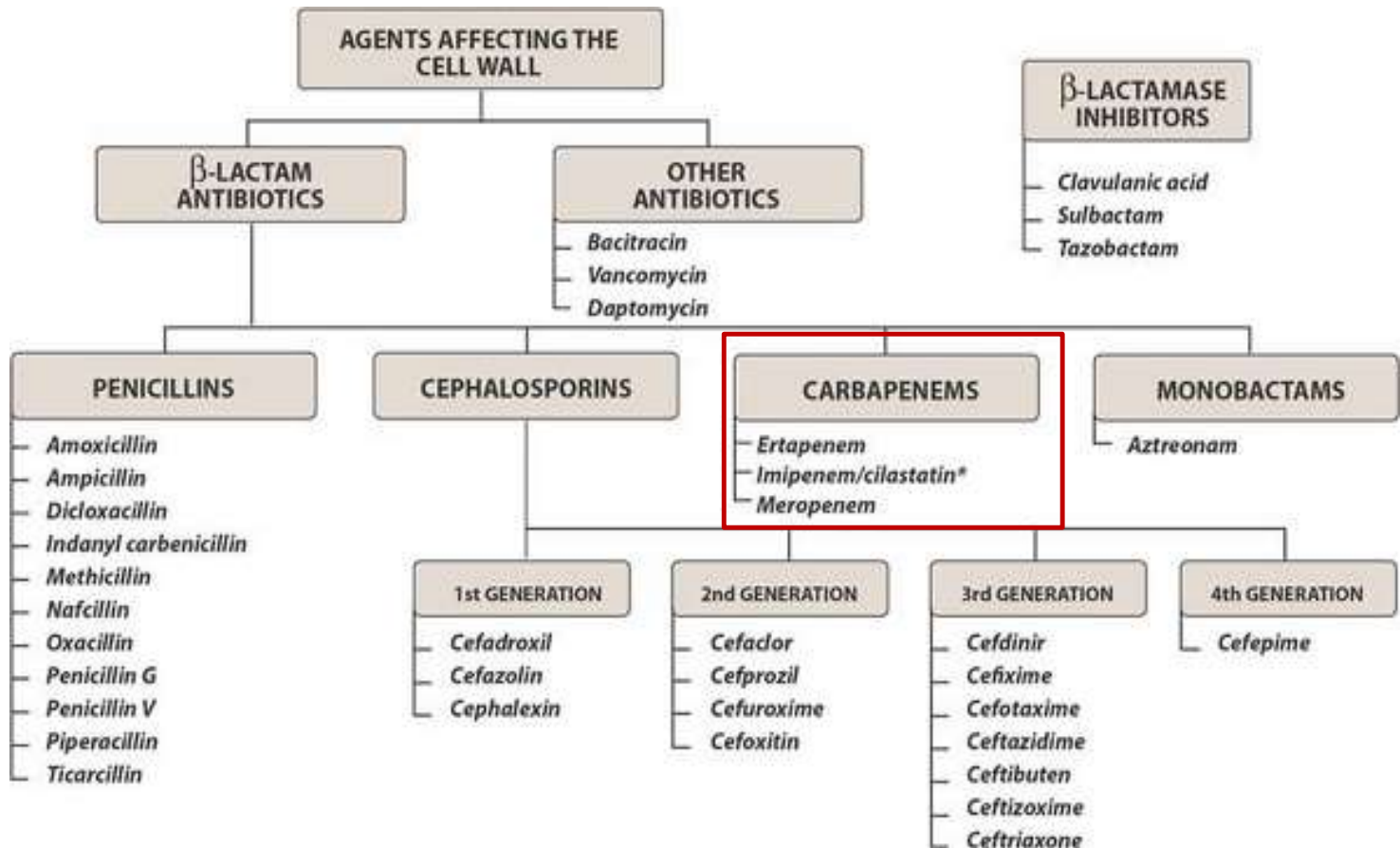
Q: Patient has allergic to Penicillins can we use Cephalosporins?

- About 5-10 % show cross-sensitivity between penicillins and cephalosporins.

- ✂ ✂ - Therefore, patients with history of anaphylaxis to penicillins **Will not**
be treated with a cephalosporin.



Bacterial cell wall synthesis inhibitors



CARD CARBAPENEMS

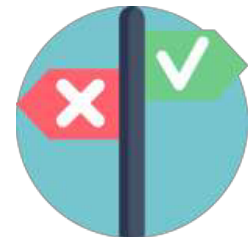
اخرستي
استخدمه
اكثره ICU

- Imipenem/ Cilastatin (I.V infusion) (GPCs)
 - Meropenem (I.V bolus or infusion) (GNRs)
 - Ertapenem (I.V)
- β -lactam antibiotics (similar to penicillins and cephalosporins)- (bind to PBP*)
 - Act by bacterial cell wall synthesis inhibition
 - They have wide activity against aerobic and anaerobic G+ and G- organisms (including Pseudomonal infections, Enterobacter, Citrobacter (*Infant meningitis*) and Serratia species). [**ACTIVE AGAINST**

ALMOST EVERYTHINGS] Last resort

- Pass BBB

*PBP- Penicillins binding protein



CARBAPENEMS: Pharmacokinetics

■ Imipenem / Cilastatin

- Is not absorbed orally (I.V infusion)

يُخرجها الدم



Inactivated by renal dehydropeptidase-I enzyme - administered with cilastatin (inhibitor of this enzyme), this lead to **increase the plasma levels of drug**

- Excreted primarily by kidney
- $T_{1/2} = 1 \text{ h}$



➤ Clinical uses:

- Used in empiric therapy for infections that required multiple antibiotics (**Nosocomial infections**)
- **MRSA are resistant.**



➤ Adverse effects

② CNS toxicity at high plasma levels (confusion, seizures)

✓ Meropenem : افعلل

- Similar to Imipenem but not metabolized by renal dehydropeptidase-I and less likely to cause seizures.

✓ Meropenem better than Imipenem

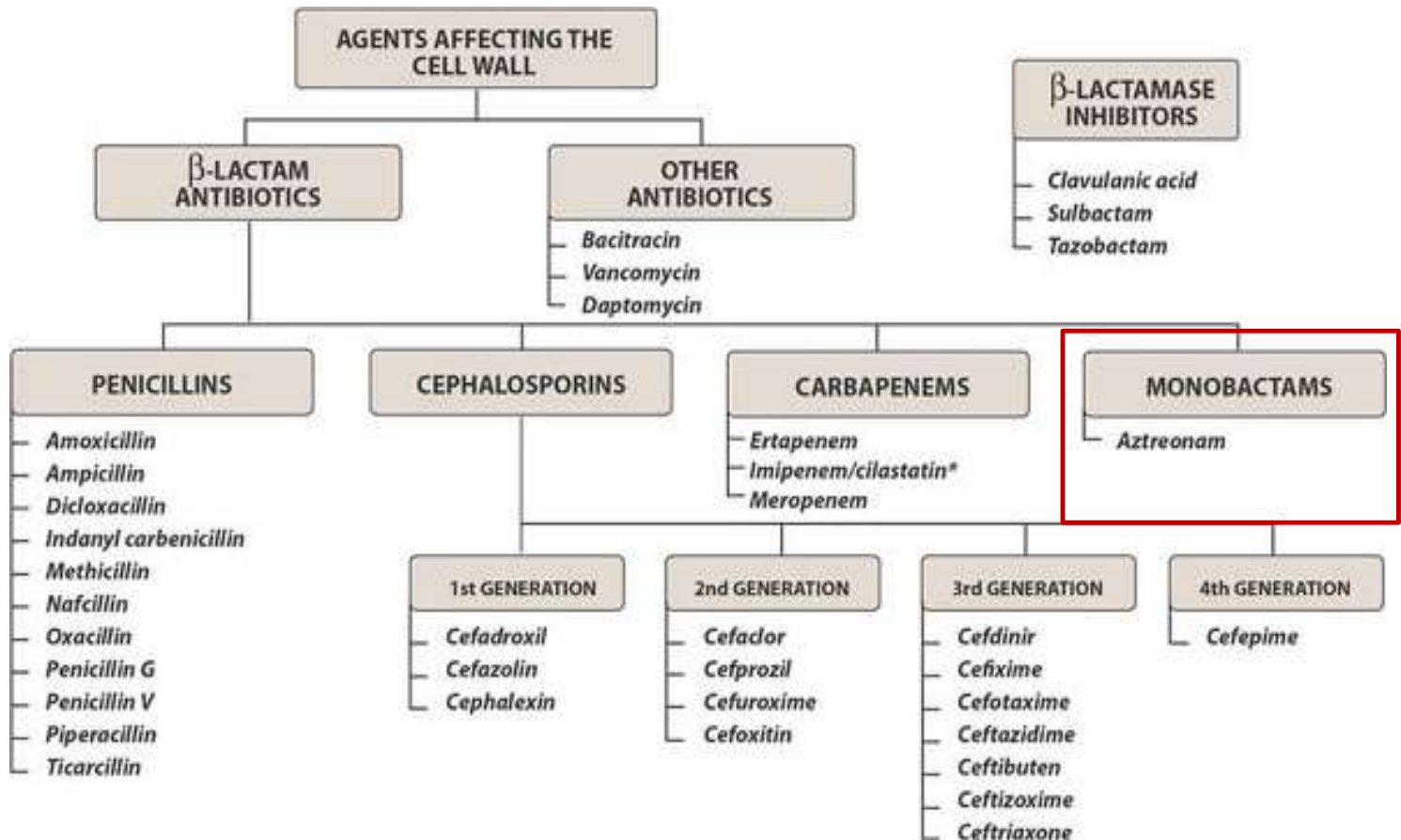


افعلل

Q: Can patient allergic to Penicillins and Cephalosporins use Carbapenems?

✗ Patients allergic to penicillins and cephalosporins may be allergic to Carbapenems

Cross-sensitivity



MONOBACTAMS

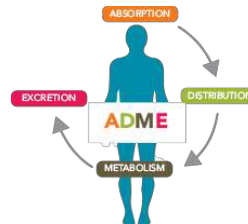
- Aztreonam (AztrEonam)

- Similar to other β -lactam drugs.
- Active against G- aerobic rods organisms (*Klebsiella*, *Pseudomonas*, *N.meningitidis*, *H.influenzae*, *enterobacteriaceae*, *N.gonorrhea* and *Serratia*)
- Not cross BBB



➤ Pharmacokinetics:

- Administered as I.V and I.M
- Limited penetration CSF
- Excreted primarily by kidney
- $T_{1/2} = 2 \text{ h}$



A E = aerobic

n= negative

EXAM

➤ Adverse effects

- Similar to other β -lactams

An elderly patient with renal impairment has a severe **Gram-negative (G-)** infection and a known allergy to both **Penicillins** and **Cephalosporins**.

Since **Aztreonam** does not show cross-reactivity with other beta-lactams

➤ Clinical uses:

MSQ

لازم السبب G-

- ✱ Alternative choice for penicillins (**Piperacillin**) and **cephalosporins** (ceftazidime) **allergic patients** for **G-** infections.
- ✱ Alternative choice to aminoglycosides, in particular, elderly renal impairments patients



Q: Can patient allergic to Penicillins and Cephalosporins use Aztreonam?

✓ Patients allergic to **all β -lactam antibiotics** can take **Aztreonam**



EXAM

What antibiotics can be used in penicillin allergy?

Most authorities recommend **avoiding cephalosporins in patients allergic to penicillins:**

G+ organisms → Consider **Macrolides** (e.g. **Erythromycin**)
or **Quinolones** (e.g. **ciprofloxacin**,
levofloxacin)
or **Vancomycin** (serious infection)

G- organisms rods → Consider **Aztreonam**



II- GLYCOPEPTIDES

- Vancomycin ; Teicoplanin
- Bacitracin

Vancomycin:

- Inhibits cell wall synthesis by inhibits transglycosylation reaction.
- Bactericidal glycoprotein
- Active against aerobic G+ organisms, especially (Staphylococci including β -lactamase producing and **MRSA**)
- Active against anaerobic G+ organism (C.difficile)
- Not effective against G- organisms because its molecule is very large to penetrate the outer membrane of G- organisms.



Historical perspective

- S.aureus $\xrightarrow[1940]{\text{Pen G}}$ Penicillin resistant S. aureus $\xrightarrow[1959]{\text{Methicillin}}$

Methicillin resistant S. aureus (MRSA)

1960-1970s

Vancomycin

Vancomycin resistant S. aureus (VRSA)

2002

**Daptomycin, linezolid,
quinupristin-dalfopristin**

?

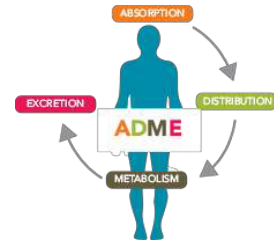


Vancomycin: pharmacokinetics



Is not absorbed from the GIT (Given orally in antibiotic-associated colitis, in particular by *C. difficile*)

- Its given I.V infusion
- Widely distributed (even CSF in inflammation)
- Excreted unchanged in the **urine**
- Dosage modification is necessary in patients with renal impairment.
- $T_{1/2} = 8 \text{ h}$



➤ Clinical uses:

- Staphylococcal infections caused by **MRSA** or **MRSE**.
- Prophylaxis for common surgical procedures in hospitals with high rates of infections due to MRSA or MRSE.
- **Antibiotics-associated colitis** (D.O.C is **Metronidazole**) Or Vancomycin orally
- Vancomycin-resistant staph then use **Daptomycin, linezolid**



EXAM

➤ Adverse effects

- Chills, Fever
- Phlebitis at the site of injection.
- Ototoxicity , Nephrotoxicity (high conc especially with aminoglycosides)
- ✱ ✱ • “**red man syndrome**” flushing (due to rapid I.V infusion- histamine release)
- Should be over at least 60 min *and slowly*



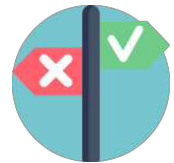
Bacitracin:



- Inhibits bacterial cell wall synthesis
- Most active against G⁺ organisms (Staphylococci including β -lactamase producing and MRSA)

➤ Clinical uses:

- Because of its nephrotoxicity, Not used systemically.
- Used topical for skin infections only.



	1 ST GENERATION	2 ND GENERATION	3 RD GENERATION	4 TH GENERATION
	Cephalexin Cephradine Cefadroxil	Cefoxitin Cefuroxime Cefaclor	Cefotaxime Cefoperazone Ceftriaxone	Cefipime
Spectrum	Mainly Gram +ve spp. and few Gram -ve bacilli	Maintain Gram +ve coverage and enhanced Gram -ve coverage	Excellent Gram -ve coverage (including <i>Pseudomonas</i>)	Wide Gram +ve and Gram -ve coverage including <i>Pseudomonas</i>
β-lactamase susceptibility	Yes	Yes	No	No
Penetration to CSF	No even in meningitis	No (except <i>cefuroxime</i>)	Yes (except <i>cefoperazone</i>)	Yes
Cross allergy with penicillin	High	Moderate	Low	Low

